



NANYANG
TECHNOLOGICAL
UNIVERSITY
SINGAPORE

SC6019 – Blockchain Privacy & Scalability

Lecture 08

Neil HAN



Blockchain Scalability

Layer 1 Solutions: On-Chain Scalability

Scaling Computation – Sharding & Blockchain Interoperability

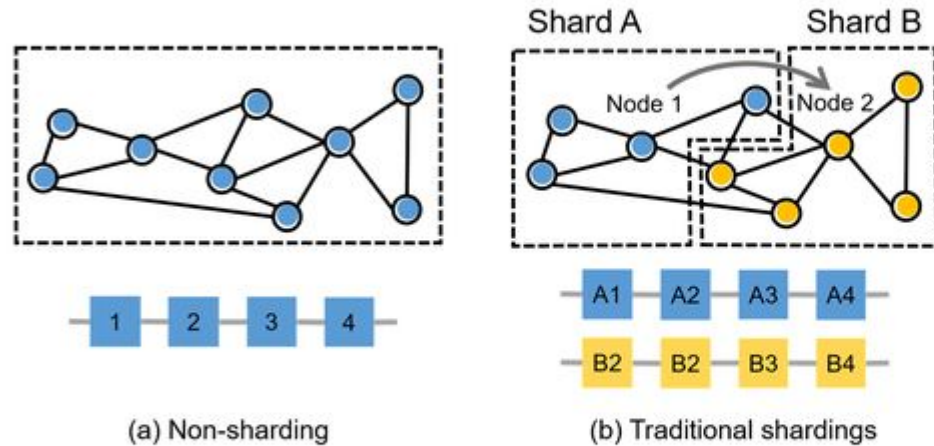
Fundamental Problem in Blockchain Scalability

"every node process all TXs, store all data, plus total ordering of TXs is required"

What if we split the TX processing into smaller groups?

Sharding Idea:

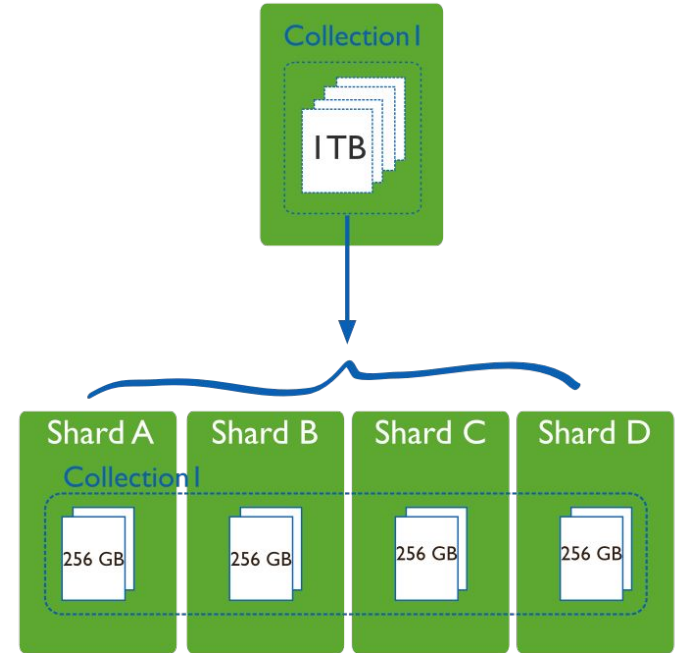
Horizontally scale by **partitioning** transaction processing into *shards*, and **parallelizing** their execution



Sharding: the concept

MongoDB:

- Huge data volume



source: <https://docs.mongodb.com/manual/sharding/>

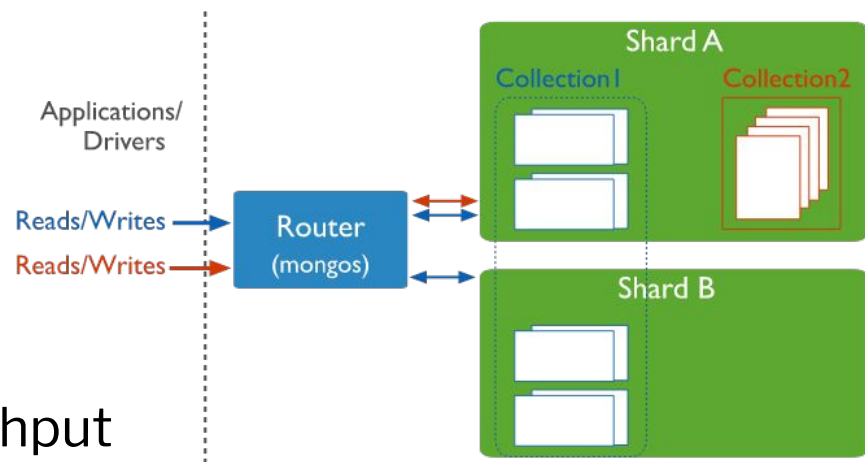
Sharding: the concept

MongoDB:

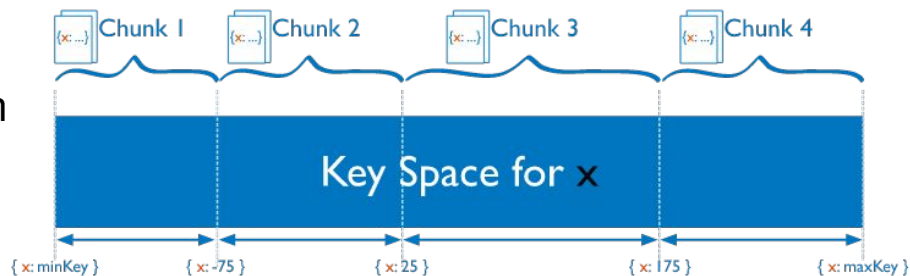
- Huge data volume
- High R/W request loads

Better parallelization => Higher throughput

Smaller search space per shard => faster response

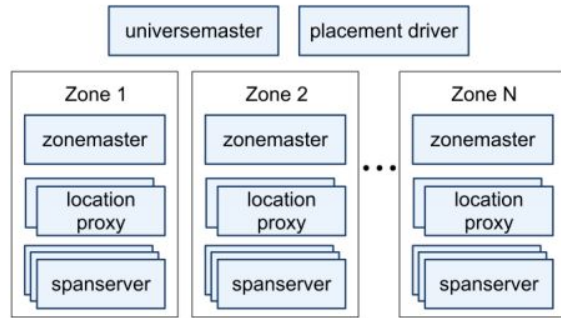


Shard key: an index field that determines which shard a piece of data/request should go



source: <https://docs.mongodb.com/manual/sharding/>

Sharding: for global distributed database/system



Partitions

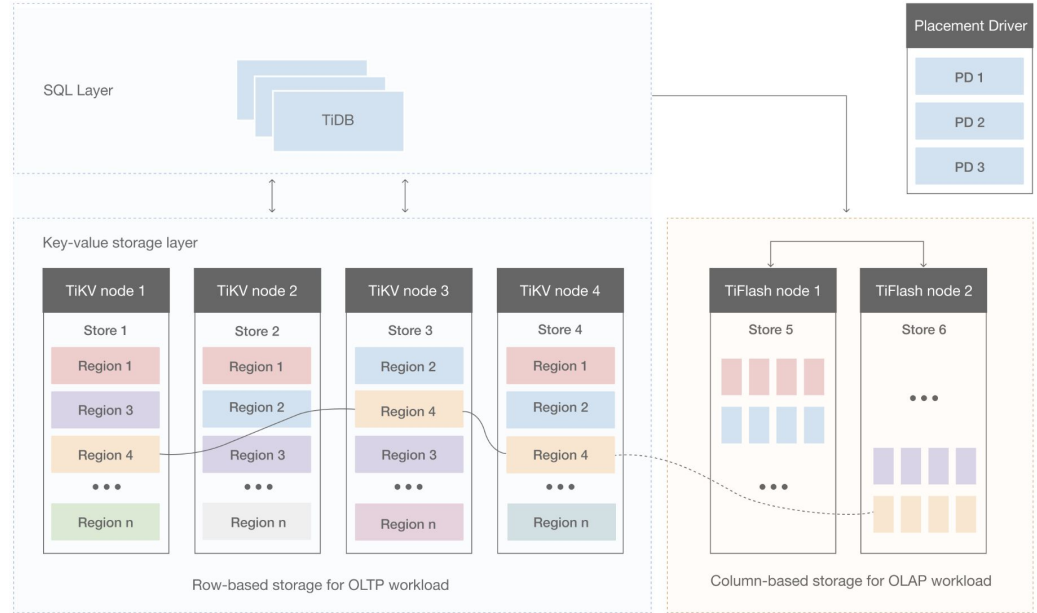
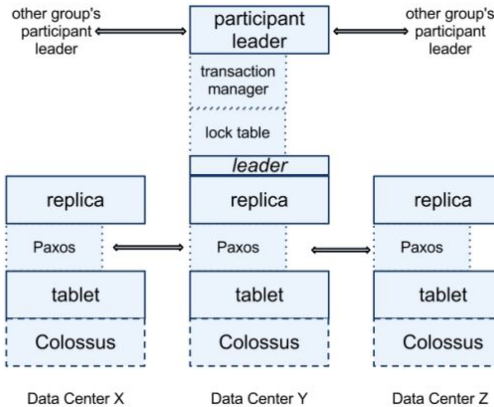
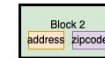


Table					
name	age	address	zipcode	city	country
alice	28	Dam Square	1111	AMS	NL
daniel	28	Java Eiland	2222	AMS	NL
jenny	30	Amstel	3333	AMS	NL
lily	24	Central	4444	AMS	NL

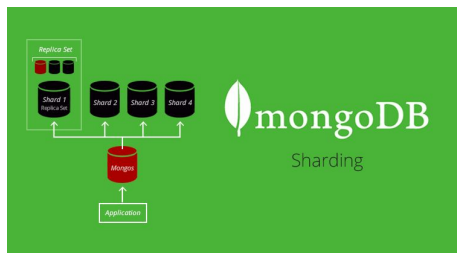


source: <https://static.googleusercontent.com/media/research.google.com/en//archive/spanner-osdi2012.pdf>

Blockchain Sharding: Challenge

Traditional sharding

- Trusted infrastructure
- Crash-failure with permissioned identity



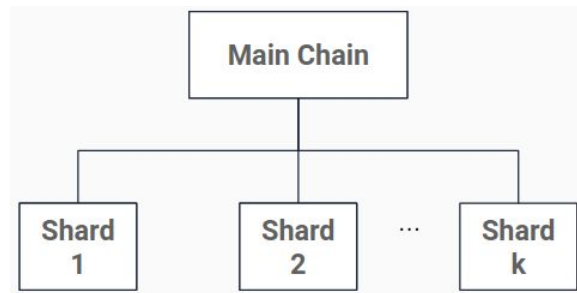
Google
Cloud
Spanner



TiDB

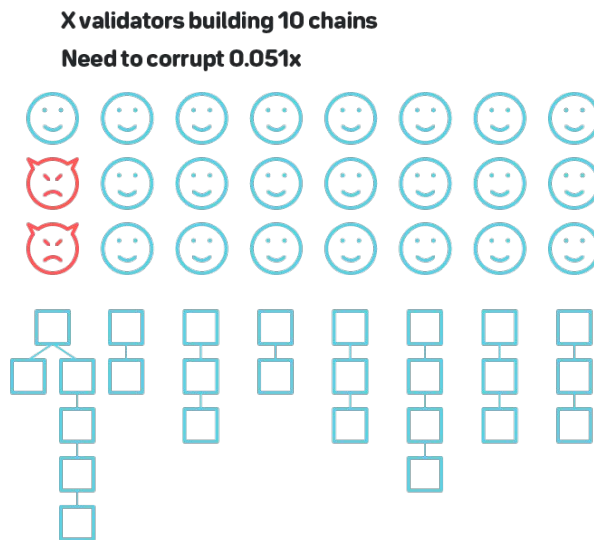
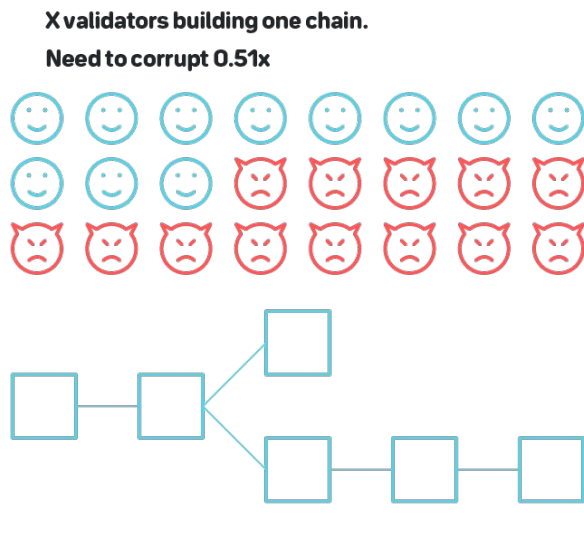
Blockchain sharding

- Adversarial network
- byzantine-failure with Sybil identity



Strawman Sharding

Each shard has their own validators and manage their chain separately



✗ Problems:

- s shard **reduces security** s folds without random validator assignment
- Can't send transaction across shard

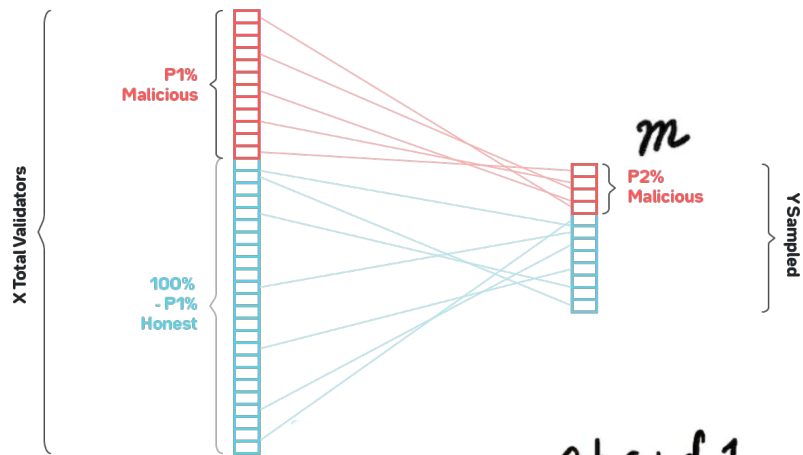
💡 Solution:

Randomly assign validators to shard

source: <https://near.org/papers/nightshade/>

What does random validator assignment do?

Let $P_1 = 30\%$



Probability a shard having more than 30%

$$Pr[P_2 > 30\%] = 1 - Pr[P_2 \leq 30\%]$$

$$= 1 - \sum_{i=0}^3 Pr[m = i]$$

$$= 1 - \sum_{i=0}^3 C_{10}^i (0.3)^i \times (0.7)^{10-i}$$

$$\approx 35\%$$

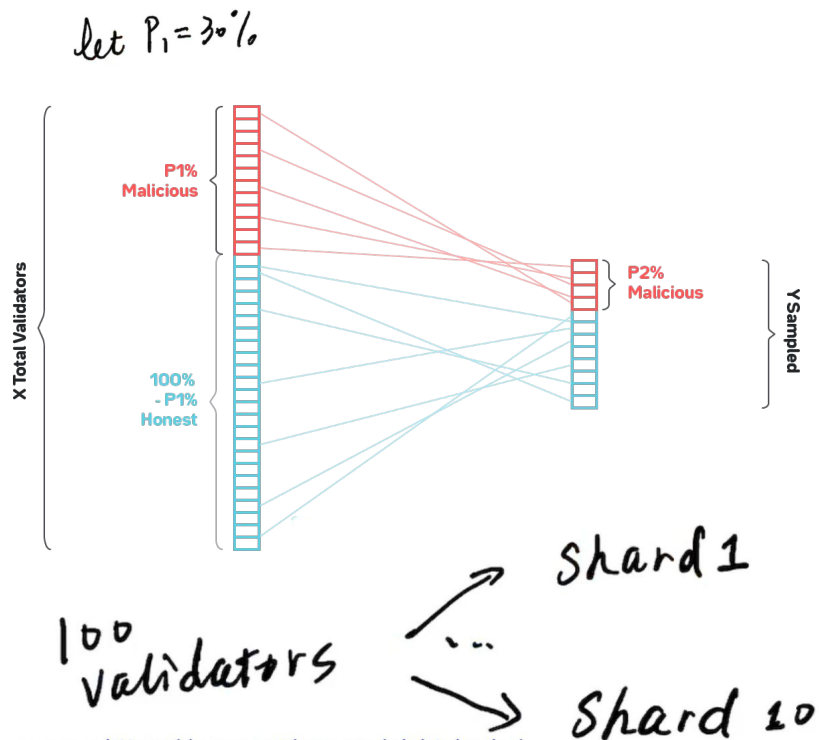
100 validators → shard 1
 ...
 → shard 10

source: <https://near.org/papers/nightshade/>

What does random validator assignment do?

Chances a shard having more than 30%

$$\Pr[P_2 > 30\%] \approx 35\%$$



source: <https://near.org/papers/nightshade/>

✓ Good:

- Doesn't reduce security much

✗ Bad:

- Still a smaller target, could be compromised after random assignment



Solution:

Randomly shuffle/rotate the validator occasionally

(needs a random beacon chain!)

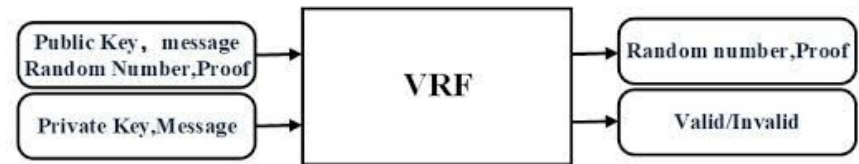
How do we get unbiased randomness?

RandHound/RandHerd¹:

"large-scale distributed protocols ... which provides *publicly-verifiable*, *unpredictable*, and *unbiasable* randomness against *Byzantine adversaries*" - Participants jointly generate

Verifiable Random Function² (VRF):

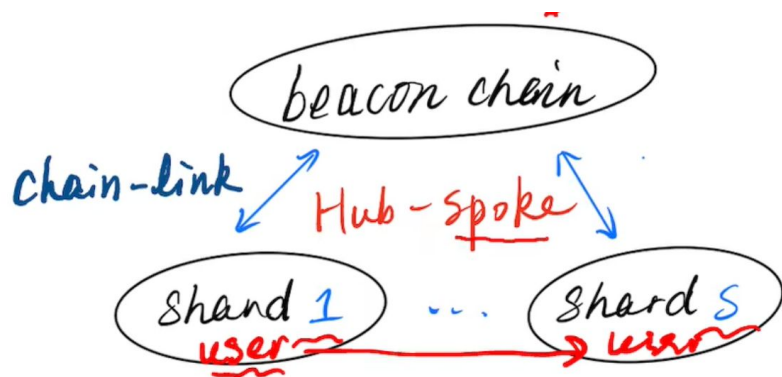
"public-key version of a keyed cryptographic hash. Only the holder of the private key can compute the hash, but anyone with public key can verify the correctness of the hash"



[1] : <https://www.ieee-security.org/TC/SP2017/papers/413.pdf>,

[2] : https://people.csail.mit.edu/silvio/Selected%20Scientific%20Papers/Pseudo%20Randomness/Verifiable_Random_Functions.pdf

How to facilitate cross-shard transaction?



! beacon chain needs updated states => validators on each shard periodically send/commit its new block (a.k.a. "**chain-link**") to beacon chain

? what to commit? Entire block or just the block header from each shard?

? where does the cross-shard transaction get validated/processed? In beacon chain or shard chain?

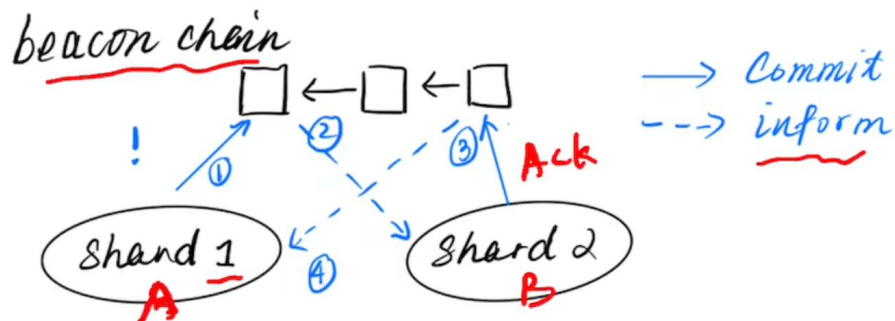
Chain-link = entire block: **transaction sharding** only

Global states in beacon chain, cross-shard transaction are *synchronously* executed in the beacon chain

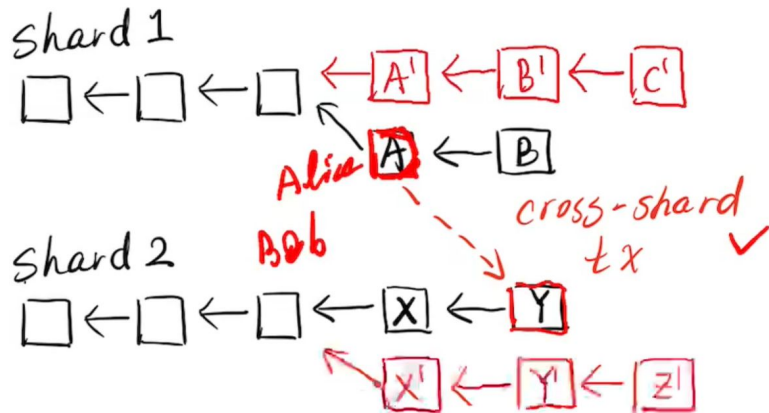
Chain-link = block header: transaction & **state sharding**

Data (namely all blockchain states) are also sharded, no global state, cross-shard transaction take effect *asynchronously* and locally inside a shard with the help of relaying via the beacon chain

Asynchronous cross-shard transaction



! Asynchronous, need **atomicity**
i.e. "train-hotel problem"



Atomicity: all-or-nothing guarantee

Either both train ticket and hotel reservation or none

(details of how to achieve atomicity omitted due to complexity)

Sharding in a Nutshell

! content in cross-link are treated as raw data blob and merkelized without validation by the beacon chain

Provide randomness,
facilitate cross-shard tx,
aggregate cross-link into finalized
block

? Cross-link:
block header or
entire block?

Shard Chain:
Run efficient BFT consensus

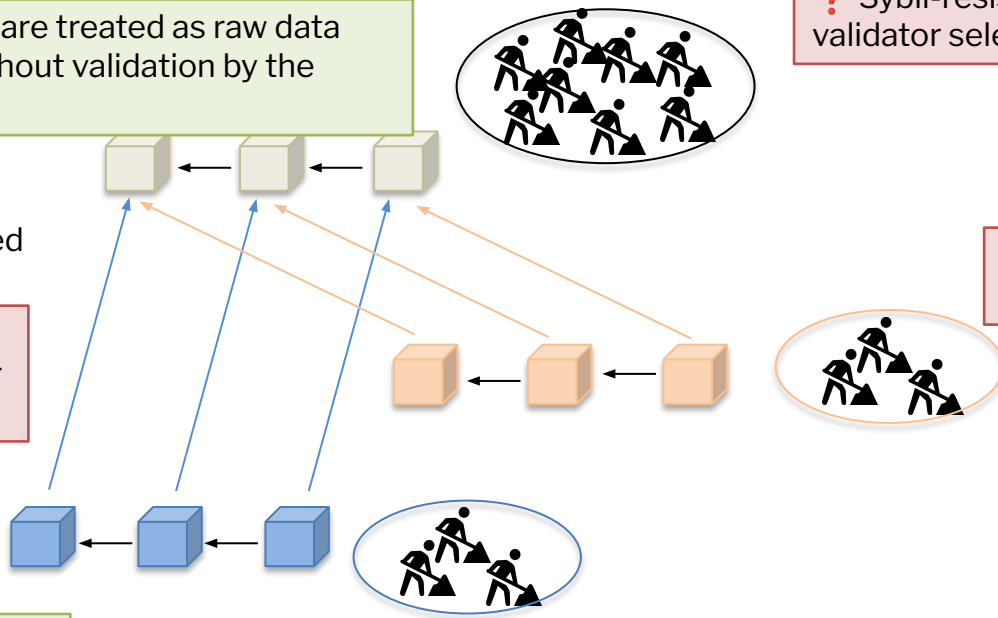
! **shard key** for incoming tx
is usually last few digits of
the address of the user
(e.g. 0x...01 in shard 1, ...
0x...09 in shard 9)

! at most a few thousands
validators in BFT !

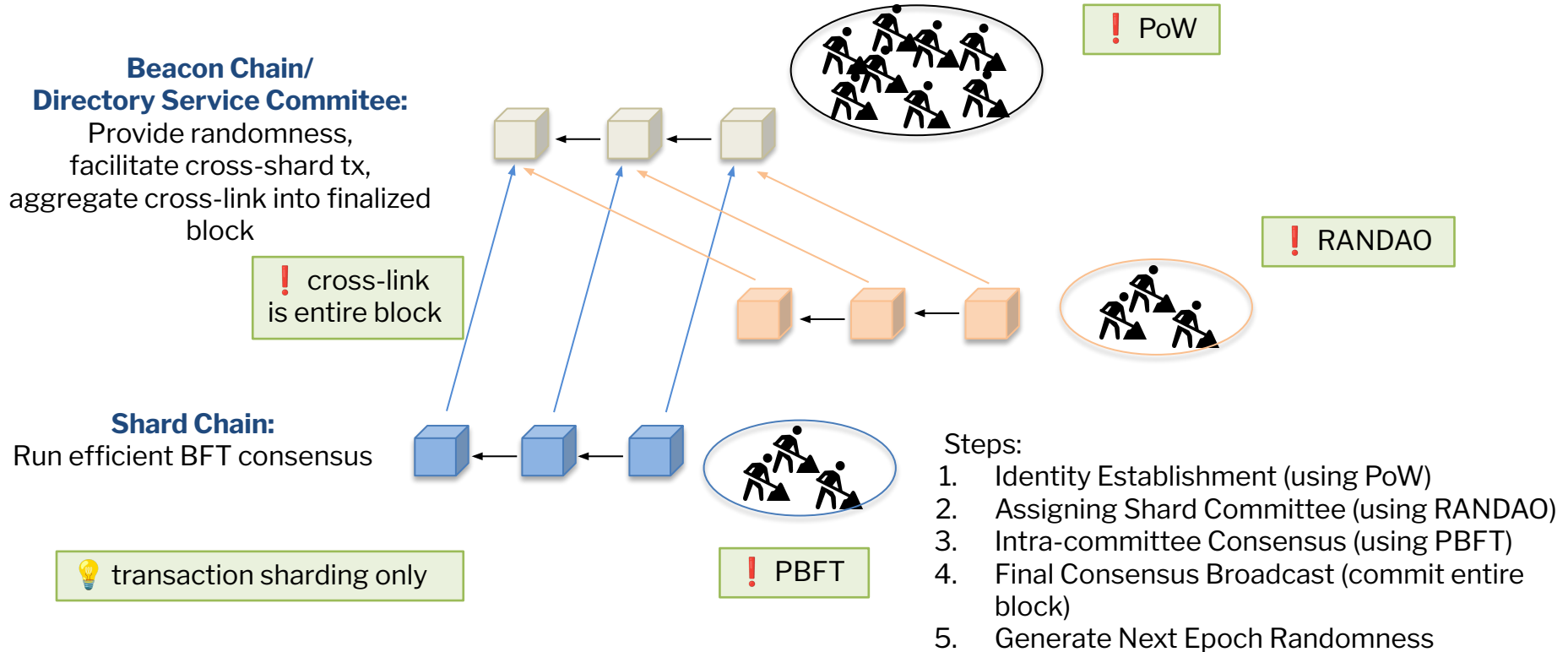
? Sybil-resistant
validator selection?

? random validator
shuffling method?

*Places where different
sharding protocols differ*

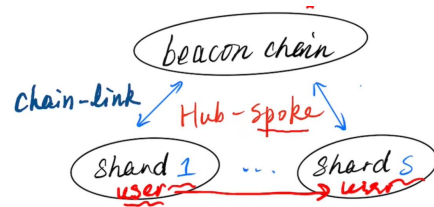


Elastico¹ [Luu et.al] / Zilliqa² Sharding



[1] : <https://loiluu.com/papers/elastico.pdf>, [2] : <https://www.zilliqa.com/>

Pause & Ponder



- What computation does validators on the beacon chain do?

Validate signatures on the shard blocks approved by the correct shard committee + process cross-shard transactions + aggregate/merkelize all raw blocks into a global state tree

- Does a validator on beacon chain require more hardware resources than that of a shard chain?

Yes. With s shards, beacon chain validators are storing all raw block from all shards, thus requires higher communication bandwidth, s times the storage, but since validators don't need to re-process these shard blocks, they doesn't necessarily require much more powerful computation. (the shard committee is the one that does the heavy work of executing and validating the shard block, while the beacon chain validator mainly does coordination and verification of proofs/commitments.)

- Why state validity and data availability are guaranteed?

Data availability: all states are globally aggregated at beacon chain level, so no issue

State validity: PBFT at shard chain level ensures correctness of each shard block, even if a shard is corrupted, users could challenge the (malicious) shard validators to provide proof of state transition at the beacon chain.

NearProtocol¹ Sharding

Beacon Chain:

Provide randomness,
facilitate cross-shard tx,
aggregate cross-link into finalized
block

! cross-link is
only block header

Shard Chain:

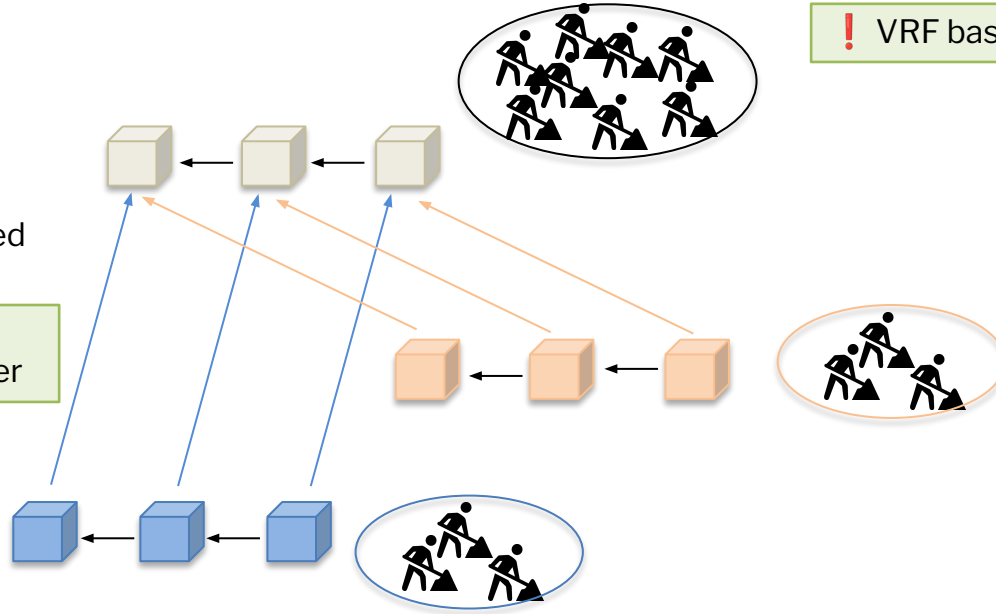
Run efficient BFT consensus

💡 transaction & state sharding

! Threshold PoS + TxFlow
(easier for low resource nodes)

! VRF based on stake

! VRF



[1]: <https://near.org/>

Pause & Ponder

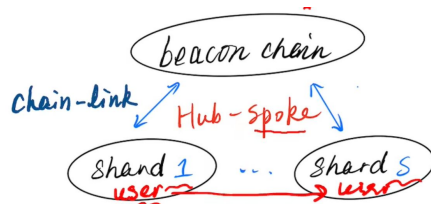
- Can we add infinite amount of shards to the system?

No, beacon chain is itself a blockchain and computationally bounded by the nodes that maintain it. But there's a multiplicative effect if any improvements made to the nodes. If each node is 4 times as fast, then the throughput of a shard chain is 4X, and the number of shards the system can accommodate is ~4X, thus in total ~16X improvement. Thus the name -- "**quadratic sharding**"

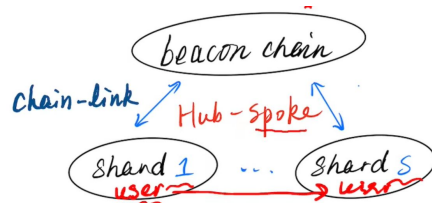
- How to ensure data availability?

This is a much harder question, to be answered in later section

- How to ensure state validity?



Ensuring State Validity, Overview



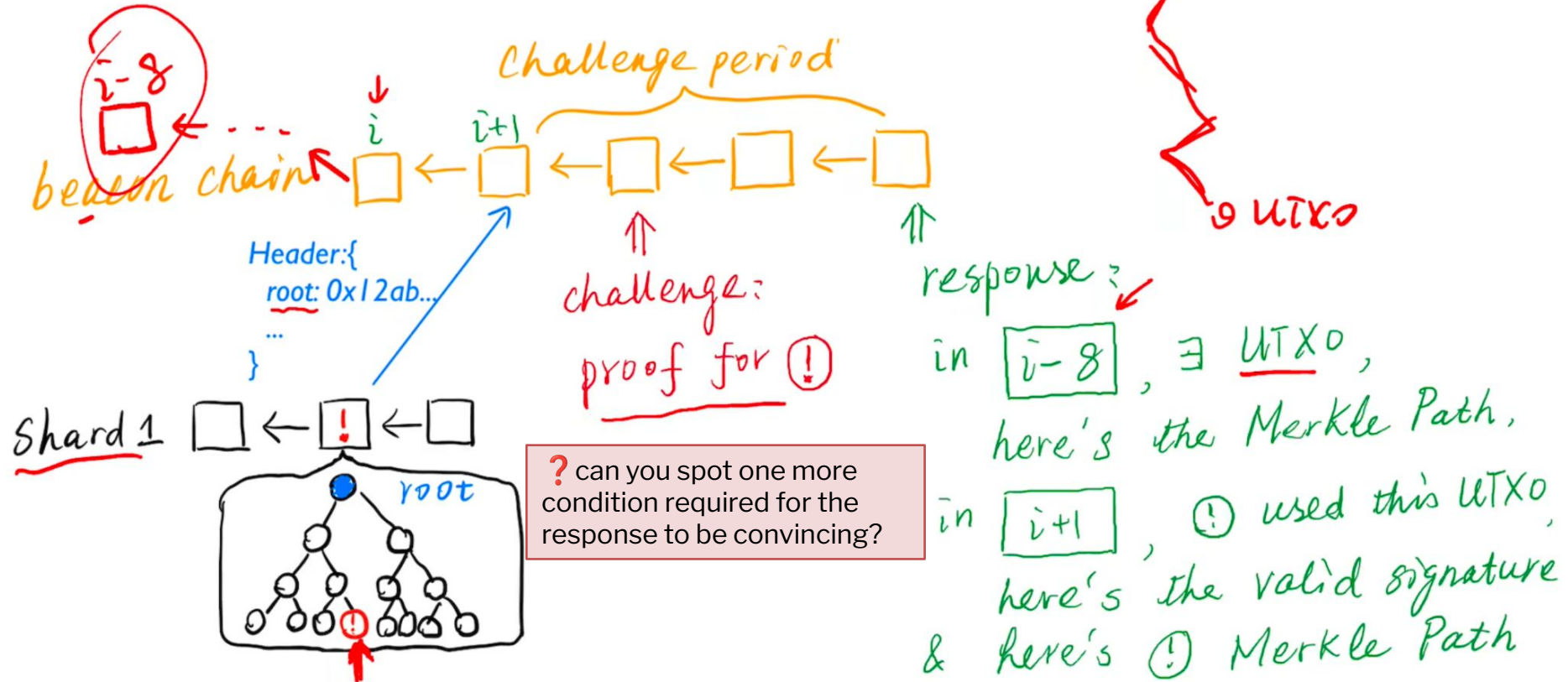
Approach 1: interactive proof of computation (aka "**Fisherman**" or "**fraud proof**")

Idea: before a predefined challenge period elapsed, anyone can question the computation integrity at certain locality; one that computes the result must provide an evidence in time, otherwise failure to respond is used as a proof of fraud

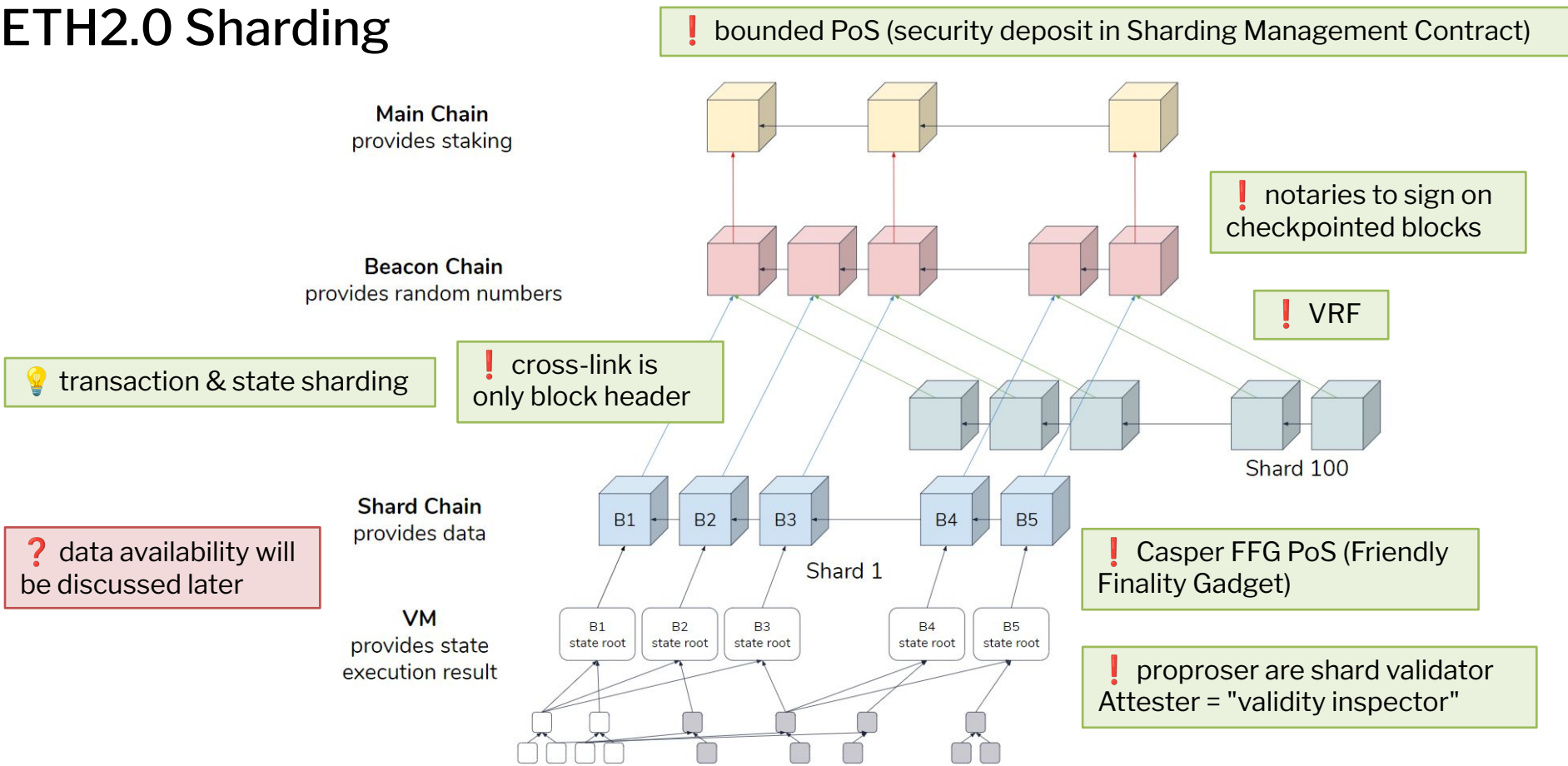
Approach 2: non-interactive cryptographic proof of computation (e.g. SNARK)

Idea: "The result of $f_1(f_2(f_3(\dots f_k(x) \dots)))$ is y , and here's a publicly-verifiable proof π

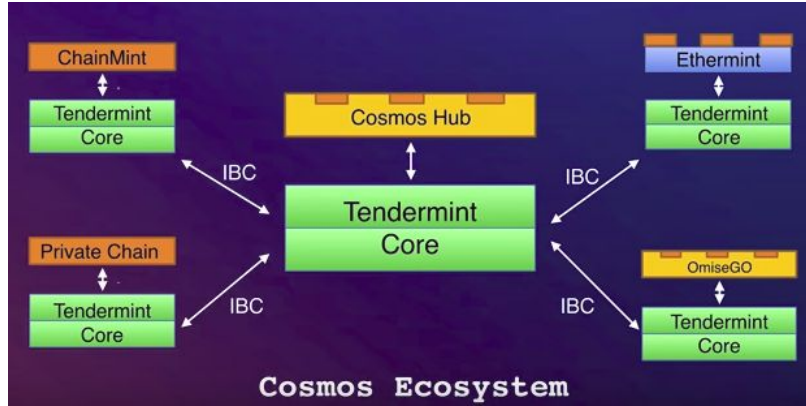
Ensuring State Validity – Fisherman



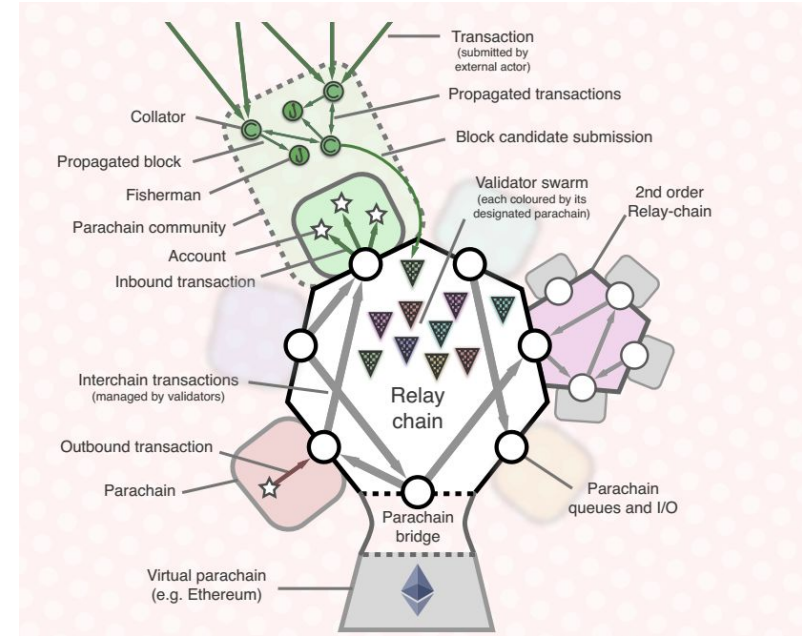
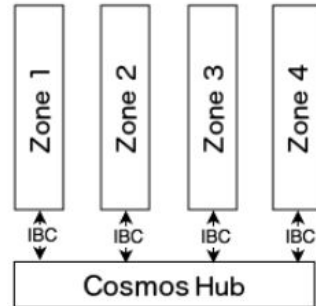
ETH2.0 Sharding



Blockchain Interoperability $\approx$ State Sharding



CØSMOS



Polkadot.

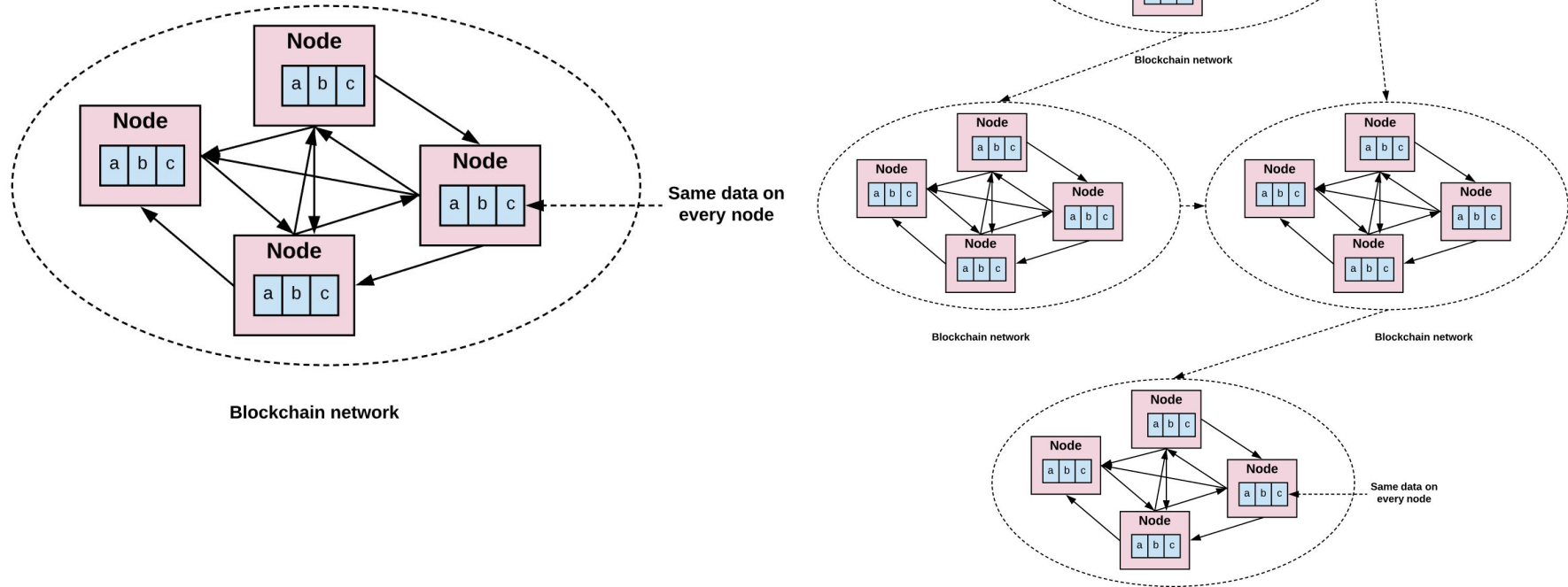
source: <https://cosmos.network/resources/whitepaper>, <https://polkadot.network/PolkaDotPaper.pdf>

Blockchain Interoperability $\approx$ State Sharding

State Sharding	Cosmos	Polkadot
Beacon chain	Cosmos Hub	Relay Chain
Shard chain	Cosmos Zone	Para Chain
All shards run BFT or PoS variants	All use Tendermint Consensus OR use any consensus then bridge to Tendermint	Pluggable consensus
Cross-shard txn	Inter-blockchain Communication (IBC)	Cross-chain txn

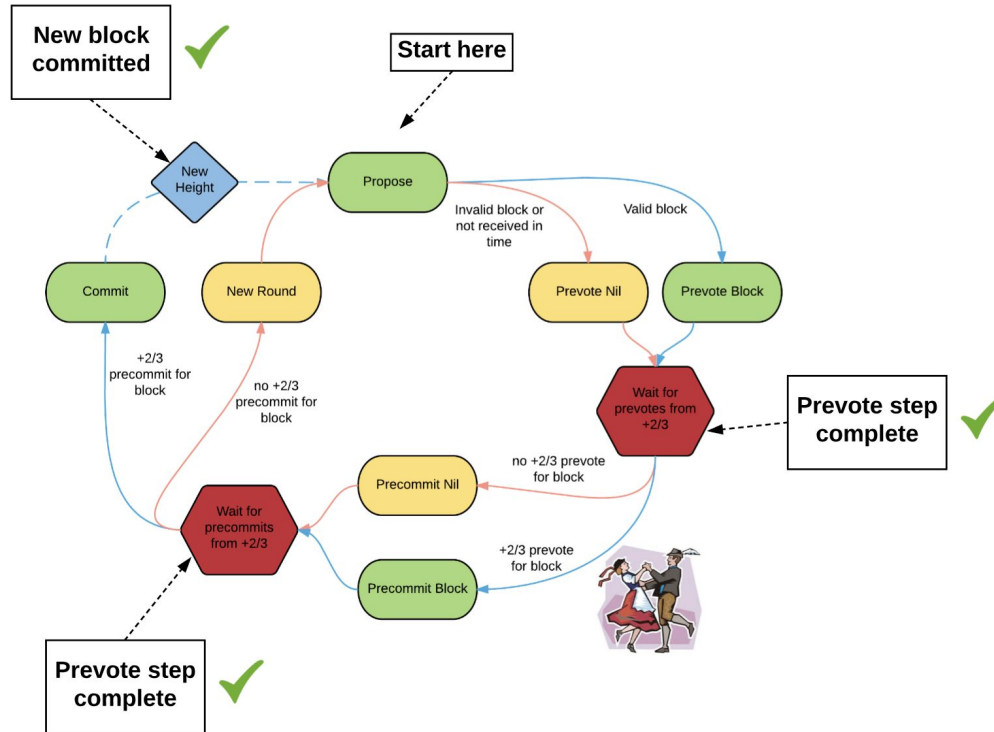
source: <https://forum.cosmos.network/t/polkadot-vs-cosmos/1397>

Cosmos



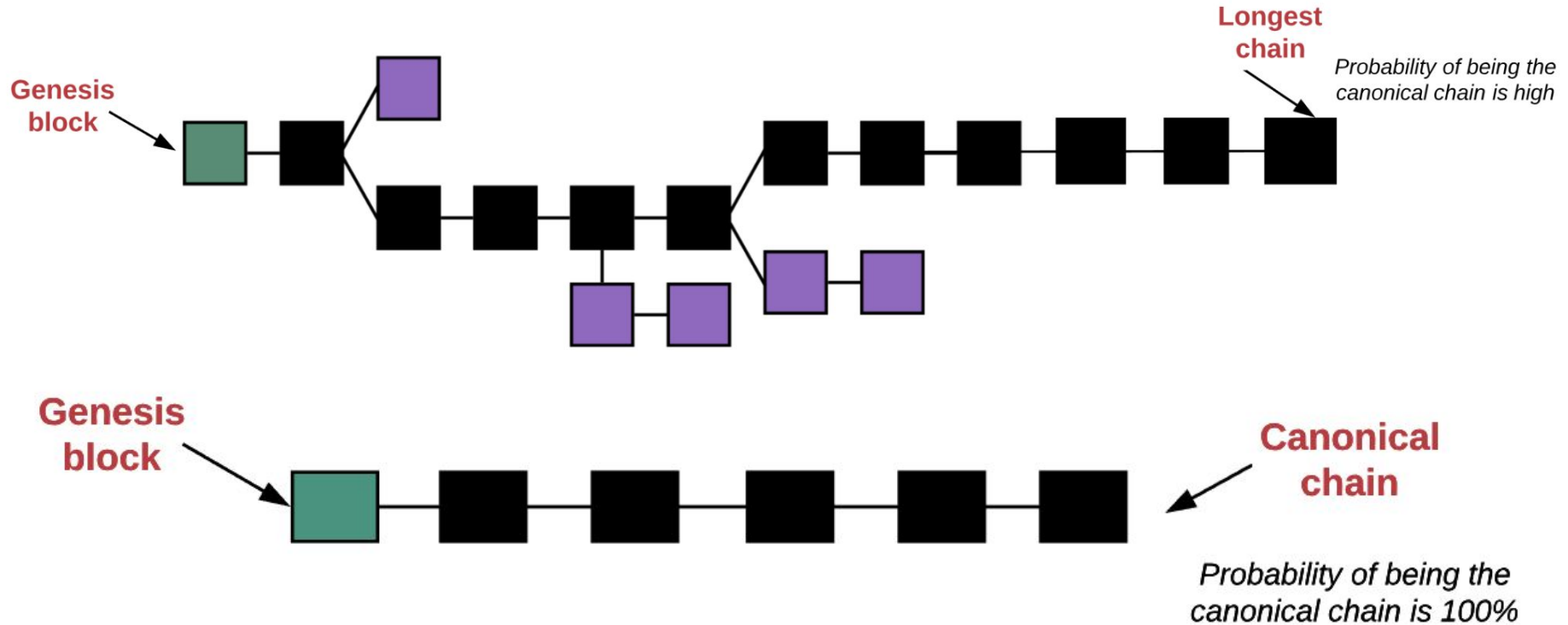
source: <https://www.preethikasireddy.com/post/how-does-cosmos-work-how-does-it-compare-to-bitcoin-and-ethereum-part-1>

Tendermint consensus



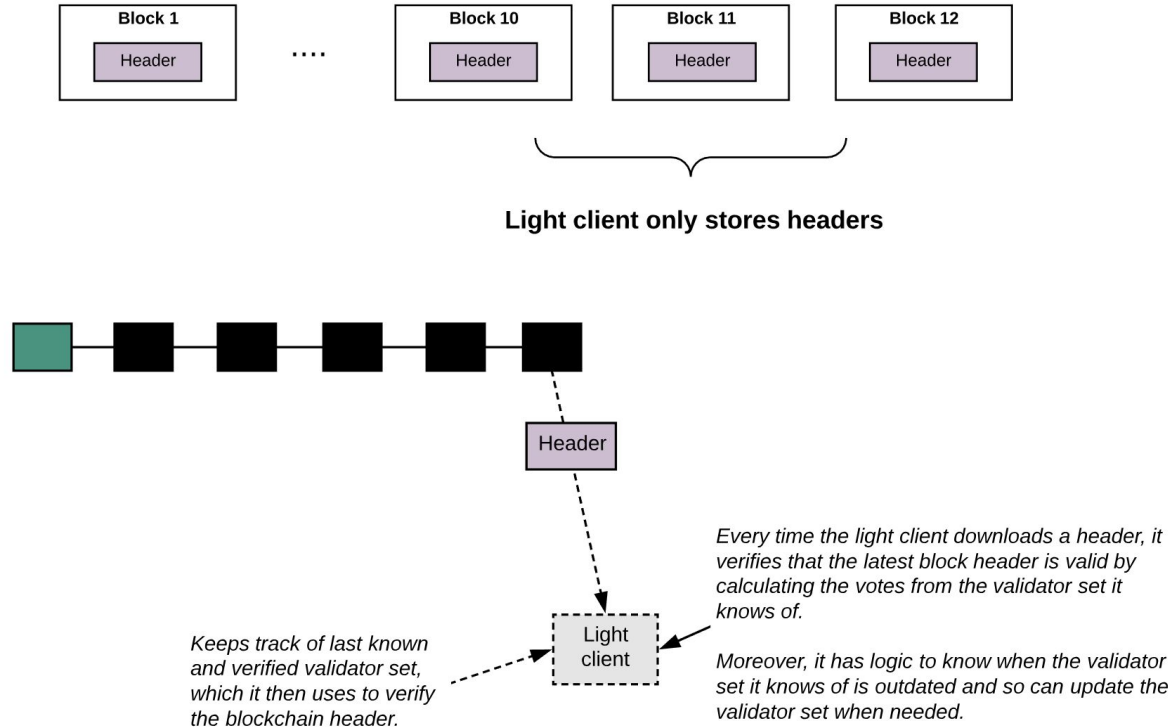
source: <https://www.preethikasireddy.com/post/how-does-cosmos-work-how-does-it-compare-to-bitcoin-and-ethereum-part-1>

Cosmos: Deterministic vs. Probabilistic



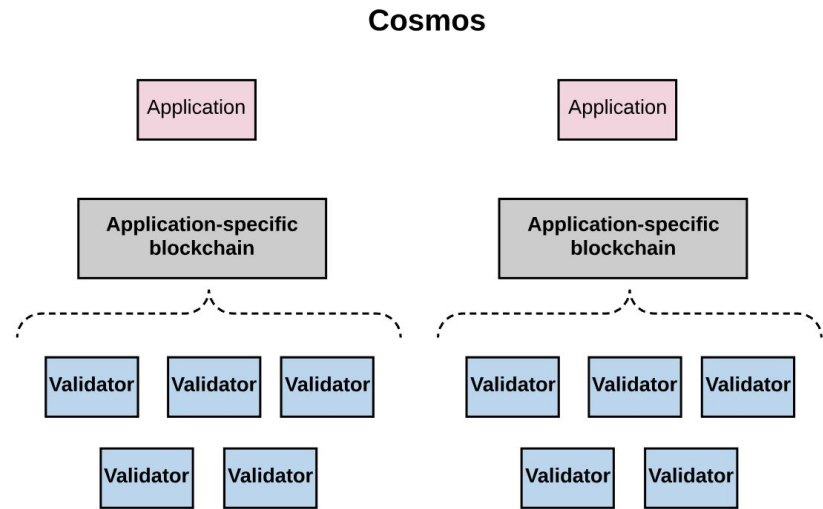
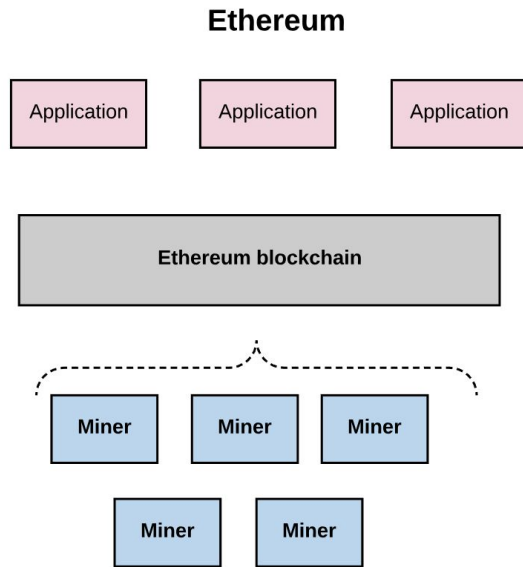
source: <https://www.preethikasireddy.com/post/how-does-cosmos-work-how-does-it-compare-to-bitcoin-and-ethereum-part-1>

Cosmos: Easier light clients



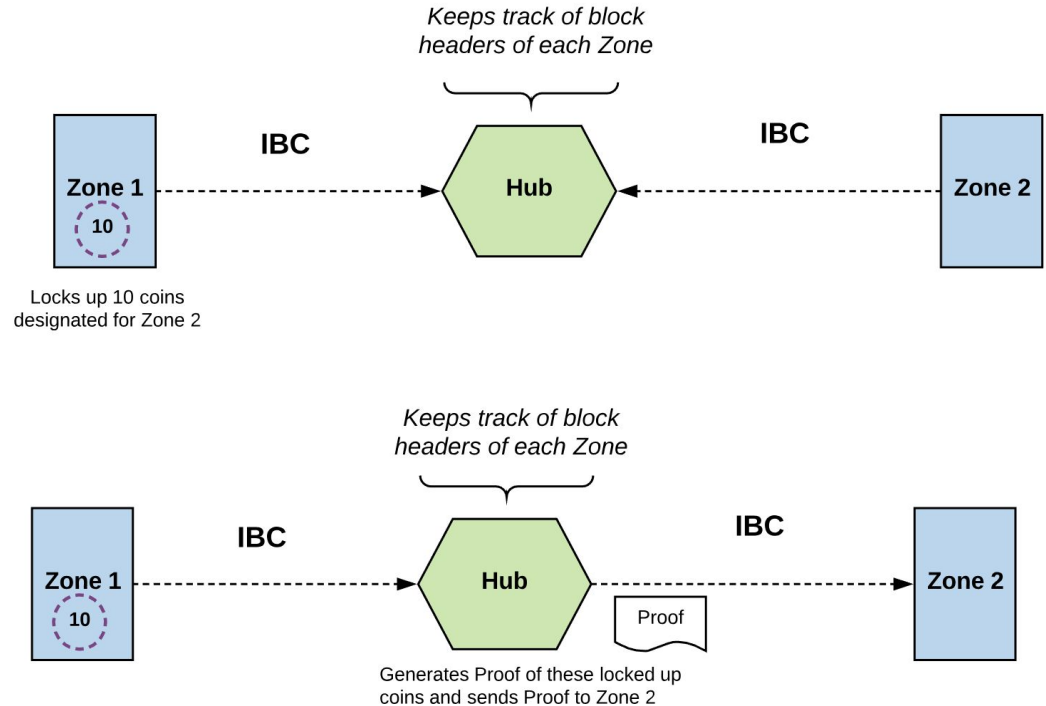
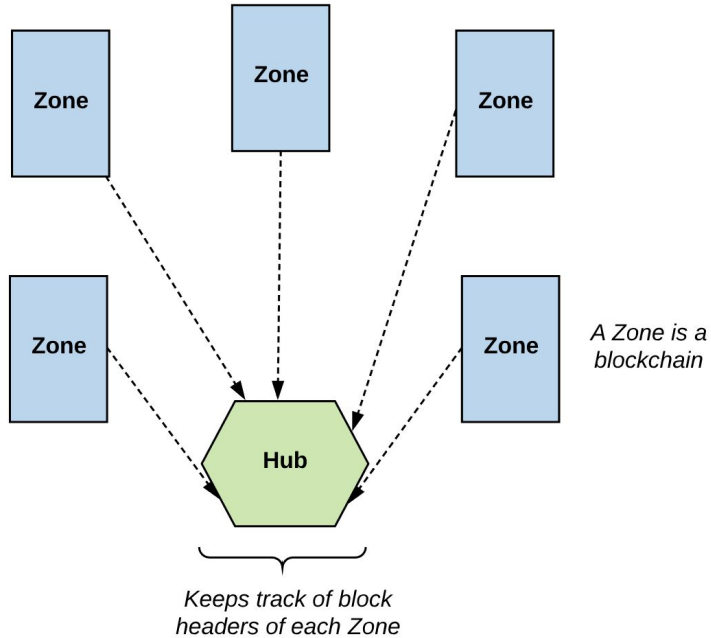
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Ethereum vs. Cosmos



source: <https://www.preethikasireddy.com/post/how-does-cosmos-work-how-does-it-compare-to-bitcoin-and-ethereum-part-1>

Hubs and Zones



source: <https://www.preethikasireddy.com/post/how-does-cosmos-work-how-does-it-compare-to-bitcoin-and-ethereum-part-1>